

Dual Representatives of Mean-Force Equilibrium

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We formulate a theory of dual representatives for mean-force equilibrium in which asymptotic control is exchanged by discrete transformations on representative labels. The first transformation is a strong-weak involution acting on the crossover coordinate $y = \log \chi$, yielding $\chi \mapsto \chi^{-1}$ and, in single-channel scaling, $g^\vee = g_*^2/g$. The second is an Z_2 orientation gauge on the mean-force eigenaxis, inducing an affine reflection on effective temperature representatives, $\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}$. We prove the involutive and affine identities, state the admissible observable class for asymptotic exchange, and derive representative constraints for effective-temperature parameterisations. Only after the formal theory is established do we instantiate it in a spin-boson qubit as Example 1, where the $g \rightarrow \infty$ sector is analytically easy in the influence-operator representation e^S , and is used to access hard low-temperature control through the dual map. We conclude with a minimal validation protocol centered on a single $y = \log \chi$ figure and identity checks.

INTRODUCTION

Dualities are a recurring organizing principle across physics: they connect descriptions that look different but encode the same underlying content, often exchanging weak and strong control regimes. The reduced equilibrium state of an open quantum system is generated by the Hamiltonian of mean force. Starting from a full Hamiltonian $H_{\text{tot}} = H_Q + H_X + H_{\text{int}}$, the reduced Gibbs operator is $\bar{\rho}_Q(\beta) = \text{Tr}_X e^{-\beta H_{\text{tot}}}$, and the exact mean-force generator is defined by $e^{-\beta H_{\text{MF}}} \propto \bar{\rho}_Q$. In finite-coupling regimes this object exists exactly but is often difficult to use in practice because asymptotic control depends strongly on which branch of parameter space is being probed. The problem addressed here is therefore not existence of the mean-force generator, but control-regime transfer: how to convert hard asymptotics into tractable ones without changing the underlying analytic family [1–8].

The key structural move is to factor the reduced operator into a bare thermal part and an influence part. This makes transparent which ingredients are universal (bare Gibbs geometry) and which carry bath-dressing information (the influence operator). Once that partition is explicit, two discrete representative actions become natural rather than ad hoc.

In this context, the most sought-after target is typically the ground-state sector ($\beta \rightarrow \infty$), which is often numerically expensive to access directly.

We isolate two discrete structures. First, a strong-weak involution acts on the crossover coordinate $y = \log \chi$, exchanging χ and χ^{-1} . Second, an orientation gauge on the mean-force eigenaxis induces an affine reflection on effective-temperature representatives, $\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}$. Together they define a duality geometry for exchanging asymptotic control sectors. Operationally, this maps a hard low-temperature/ground-state computation to a numerically trivial dual representative regime.

The manuscript is theory-first. Sections 2–5 establish the formal structure independently of spin-boson details.

The spin-boson qubit appears only as Example 1, where closed formulas make the transfer mechanism explicit and computationally testable.

The main non-claim is pointwise equality under representative inversion. The results concern asymptotic exchange of control, not parameter-level physical equivalence. A minimal, theory-led validation protocol is specified later in the manuscript.

Relative to the previous closure manuscript [9], this paper shifts emphasis from model-first derivation to representative-level duality structure, while keeping explicit scope separation from full spin-boson universality analyses [10–13].

GENERAL DUALITY FRAMEWORK

Setup from the full Hamiltonian. Take a subsystem Q and environment X with

$$H_{\text{tot}} = H_Q + H_X + H_{\text{int}}, \quad H_{\text{int}} = g f \otimes B. \quad (1)$$

Here g is a dimensionless coupling amplitude, f is a system operator, and B is a bath operator (with bath mean absorbed so $\text{Tr}_X(Be^{-\beta H_X}) = 0$). Define

$$Z_X(\beta) := \text{Tr}_X e^{-\beta H_X}, \quad Z_Q(\beta) := \text{Tr}_Q e^{-\beta H_Q}, \quad (2)$$

$$\rho_Q^{\text{th}}(\beta) := Z_Q(\beta)^{-1} e^{-\beta H_Q}. \quad (3)$$

The unnormalized reduced equilibrium operator is

$$\bar{\rho}_Q(\beta) := \text{Tr}_X e^{-\beta H_{\text{tot}}}, \quad (4)$$

with normalized state $\rho_Q = \bar{\rho}_Q / \text{Tr}_Q \bar{\rho}_Q$.

The Hamiltonian of mean force is defined exactly by

$$e^{-\beta H_{\text{MF}}(\beta)} := \frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)}. \quad (5)$$

This identity is purely definitional and non-perturbative [1–4].

Definition 1 (Influence operator). Define S_β by

$$e^{S_\beta} := e^{+\beta H_Q/2} \frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} e^{+\beta H_Q/2}. \quad (6)$$

Equivalently,

$$\frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} = e^{-\beta H_Q/2} e^{S_\beta} e^{-\beta H_Q/2}. \quad (7)$$

Hence

$$\rho_Q(\beta) = \frac{Z_Q(\beta)}{Z_{\text{MF}}(\beta)} (\rho_Q^{\text{th}}(\beta))^{1/2} e^{S_\beta} (\rho_Q^{\text{th}}(\beta))^{1/2}, \quad (8)$$

where $Z_{\text{MF}}(\beta) := \text{Tr}_Q e^{-\beta H_{\text{MF}}(\beta)}$.

This thermal-plus-influence partition is the structural backbone used below: bare thermal factors carry the reference Gibbs geometry, while S_β carries the interaction dressing. The duality results concern representative maps acting on this dressed family. It is the operator-level equilibrium counterpart of influence-functional dressing [9, 14].

Definition 2 (Representative family). Let $\mathcal{P}$ be a parameter manifold and $\mathcal{O}$ a set of observables. A representative family is a map

$$\mathcal{R} : \mathcal{P} \rightarrow \mathcal{O}, \quad p \mapsto \{O_k(p)\}_{k \in K}, \quad (9)$$

where each O_k is evaluated from one fixed analytic functional form, while p is allowed to change by discrete representative transformations.

Definition 3 (Crossover coordinate). Assume there exists a positive scalar control coordinate χ and its logarithmic coordinate y ,

$$\chi = \chi(\beta, \lambda, \vartheta) > 0, \quad (10)$$

$$y := \log \chi. \quad (11)$$

In the single-channel setting used below,

$$\chi(\beta, g, \vartheta) = g^2 \chi_0(\beta, \vartheta), \quad \chi_0 > 0. \quad (12)$$

Definition 4 (Admissible observable class). An observable Q is admissible if it can be written as

$$Q(\beta, g, \vartheta) = F(\beta, \vartheta; \phi_1(\chi), \dots, \phi_n(\chi), u), \quad (13)$$

where F is smooth in its arguments, ϕ_j have asymptotic expansions for $\chi \rightarrow 0$ and $\chi \rightarrow \infty$, and u is an oriented scalar (possibly absent) that is odd under orientation-gauge flip.

Proposition 1 (Representative-map admissibility). *If Q belongs to the admissible class in Definition 4, then discrete maps acting on representative labels (χ, u) may exchange the computational asymptotic regime of Q while preserving the same analytic functional form F .*

Proof. The map acts only on the arguments of F , not on F itself. Hence transformed evaluations are obtained by argument substitution and remain within the same closed-form family. $\square$

Theory/data separation. The next three sections use only the structure above. Model closure and bath-specific formulas enter only in the spin-boson example section.

STRONG-WEAK INVOLUTION

Involution on the crossover coordinate. Let

$$y = \log \chi, \quad \chi > 0. \quad (14)$$

Define

$$\mathcal{D}_{\text{sw}} : y \mapsto -y \iff \chi \mapsto \chi^{-1}. \quad (15)$$

Theorem 1 (Strong-weak involution). *Assume one-channel scaling*

$$\chi(\beta, g, \vartheta) = g^2 \chi_0(\beta, \vartheta), \quad \chi_0(\beta, \vartheta) > 0. \quad (16)$$

Define the self-dual scale

$$g_\star(\beta, \vartheta) := \chi_0(\beta, \vartheta)^{-1/2}, \quad (17)$$

and the dual representative

$$g^\vee(\beta, \vartheta; g) := \frac{g_\star(\beta, \vartheta)^2}{g} = \frac{1}{\chi_0(\beta, \vartheta) g}. \quad (18)$$

Then:

$$\chi(\beta, g^\vee, \vartheta) = \chi(\beta, g, \vartheta)^{-1}, \quad (19)$$

$$(g^\vee)^\vee = g, \quad (20)$$

$$y(\beta, g^\vee, \vartheta) = -y(\beta, g, \vartheta). \quad (21)$$

Proof. By substitution,

$$\chi(\beta, g^\vee, \vartheta) = \chi_0 \left(\frac{1}{\chi_0 g} \right)^2 = \frac{1}{\chi_0 g^2} = \chi(\beta, g, \vartheta)^{-1}. \quad (22)$$

Applying the same transformation twice gives

$$(g^\vee)^\vee = \frac{1}{\chi_0 g^\vee} = \frac{1}{\chi_0} \chi_0 g = g. \quad (23)$$

Taking logarithms yields $y \mapsto -y$. $\square$

Proposition 2 (Asymptotic exchange). *Let Q be admissible in the sense of Definition 4, and suppose the χ -dependent building blocks ϕ_j have weak/strong asymptotic expansions at $\chi \rightarrow 0$ and $\chi \rightarrow \infty$. Then the strong- χ asymptotic expansion of Q at (β, g, ϑ) is obtained by evaluating the weak- χ expansion at the dual representative $(\beta, g^\vee, \vartheta)$, and conversely.*

Proof. By Theorem 1, $\chi(\beta, g^\vee, \vartheta) = \chi(\beta, g, \vartheta)^{-1}$. Therefore any weak expansion $\phi_j(\chi) \sim \sum_n a_{j,n} \chi^n$ at $g^\vee$ becomes $\sum_n a_{j,n} \chi(\beta, g, \vartheta)^{-n}$, which is a strong expansion at g . The reverse direction is identical. $\square$

Interpretation boundary. The proposition transfers asymptotic control between representatives. It is not a claim that $Q(\beta, g, \vartheta) = Q(\beta, g^\vee, \vartheta)$ pointwise.

ORIENTATION GAUGE AND AFFINE REFLECTION

Orientation representative gauge. For qubit mean-force parametrisations with longitudinal dressing scalar Σ_z , define

$$\mathcal{G} : \Sigma_z \mapsto -\Sigma_z. \quad (24)$$

This is a representative-level Z_2 choice of eigenaxis orientation.

Definition 5 (Effective-temperature representative). Let

$$a := \frac{\beta\omega_q}{2}, \quad u := \gamma(\chi)U, \quad \Theta := \operatorname{arctanh}(u) - a, \quad (25)$$

with $|u| < 1$. Define the effective-temperature representative

$$\beta_{\text{eff}} := -\frac{2}{\omega_q}\Theta = \beta - \delta\beta, \quad \delta\beta := \frac{2}{\omega_q}\operatorname{arctanh}(u). \quad (26)$$

Under $\mathcal{G}$, $U \mapsto -U$, hence $u \mapsto -u$.

Lemma 1 (Affine reflection). *Under the orientation-gauge map $\mathcal{G}$,*

$$\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}. \quad (27)$$

Proof. Since $u \mapsto -u$, one has

$$\delta\beta' = \frac{2}{\omega_q}\operatorname{arctanh}(-u) = -\delta\beta. \quad (28)$$

Therefore

$$\beta'_{\text{eff}} = \beta - \delta\beta' = \beta + \delta\beta = 2\beta - (\beta - \delta\beta) = 2\beta - \beta_{\text{eff}}. \quad (29)$$

□

Lemma 2 (Even/odd observable split). *Let $O_{\text{rad}} = F(\chi)$ with no explicit dependence on u . Then*

$$O'_{\text{rad}} = O_{\text{rad}} \quad \text{under } \mathcal{G}. \quad (30)$$

If $O_{\text{odd}} = H(\chi, u)$ with $H(\chi, -u) = -H(\chi, u)$, then

$$O'_{\text{odd}} = -O_{\text{odd}}. \quad (31)$$

For longitudinal thermal representatives, the transformation is affine through Lemma 1, not generally a sign flip.

Proof. Gauge action leaves χ invariant and flips u . The stated parity properties follow by direct substitution into F and H . The affine law for β_{eff} was proved above. □

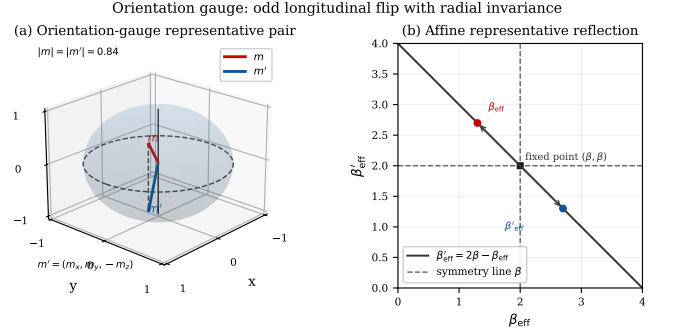


FIG. 1. Geometric action of the orientation gauge. (a) Bloch-sphere representative pair m and $m' = (m_x, m_y, -m_z)$, connected by the dashed longitudinal segment: radius is unchanged ($|m| = |m'|$) while the oriented longitudinal component flips. (b) Affine representative map for effective temperature, $\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}$, shown as reflection about the fixed point (β, β) and symmetry axes at β .

Implementation note. Figure 1 is generated by `paper_dual_hmf/validation/make_orientation_gauge_bloch_figure.py`.

COMPOSED DUALITY ACTION AND OBSERVABLE CONSTRAINTS

Composed representative action. Define

$$\mathcal{D}_{\text{comp}} := \mathcal{G} \circ \mathcal{D}_{\text{sw}}. \quad (32)$$

Proposition 3 (Composed-action structure). *The maps $\mathcal{D}_{\text{sw}}$ and $\mathcal{G}$ are both involutions on representative labels. Their action on the radial sector is always commuting:*

$$(\mathcal{G} \circ \mathcal{D}_{\text{sw}})\chi = (\mathcal{D}_{\text{sw}} \circ \mathcal{G})\chi = \chi^{-1}. \quad (33)$$

On oriented sectors, commutation holds whenever the dual map preserves orientation parity $u^\vee(-u) = -u^\vee(u)$. In that case, the generated discrete group is $Z_2 \times Z_2$.

Proof. Involution of $\mathcal{D}_{\text{sw}}$ follows from Theorem 1, and involution of $\mathcal{G}$ is immediate from sign flip. The radial statement follows because $\mathcal{G}$ acts trivially on χ . Parity preservation is the sufficient condition for commuting action on u . □

Constraint on representative fits. Suppose a saturated-regime radius is parameterised as $r = r(\beta_{\text{eff}})$. Orientation pairing then imposes

$$r' = r(2\beta - \beta_{\text{eff}}). \quad (34)$$

This gives a non-trivial consistency relation for any empirical β_{eff} -collapse.

Corollary 1 (Ratio constraint).

$$R(\beta, \theta) := \frac{r(2\beta - \beta_{\text{eff}})}{r(\beta_{\text{eff}})}. \quad (35)$$

At the symmetry point $\Theta = 0$ (equivalently $\beta_{\text{eff}} = 0$), the paired representatives are $(0, 2\beta)$, and R provides a sharp local diagnostic of representative consistency.

Temperature-driven branch exchange. The dual map also yields a direct statement about high- T /low- T control transfer when χ_0 has monotone endpoint scaling.

Corollary 2 (Classical-quantum exchange of control). Assume one-channel scaling $\chi(\beta, g, \theta) = g^2 \chi_0(\beta, \theta)$ with $g > 0$, and suppose

$$\chi_0(\beta, \theta) \rightarrow 0 \quad (\beta \rightarrow 0), \quad \chi_0(\beta, \theta) \rightarrow \infty \quad (\beta \rightarrow \infty). \quad (36)$$

Then, at fixed g ,

$$\beta \rightarrow 0 : \chi(\beta, g, \theta) \rightarrow 0, \quad g^\vee(\beta, \theta; g) \rightarrow \infty, \quad (37)$$

$$\beta \rightarrow \infty : \chi(\beta, g, \theta) \rightarrow \infty, \quad g^\vee(\beta, \theta; g) \rightarrow 0. \quad (38)$$

Hence low-temperature strong- χ asymptotics at g are transferred to weak- χ asymptotics at $g^\vee$, while high-temperature weak- χ asymptotics map to a strong-coupling dual representative.

Proof. From Theorem 1, $g^\vee(\beta, \theta; g) = 1/(\chi_0(\beta, \theta)g)$ and $\chi(\beta, g^\vee, \theta) = \chi(\beta, g, \theta)^{-1}$. The endpoint limits follow by substitution of the assumed χ_0 scaling. $\square$

NORMAL DUALITY AND GEOMETRIZATION

Question. The fixed- β dual map $g \mapsto g^\vee = g_\star^2/g$ reflects the scalar coordinate $y = \log \chi$ but keeps β unchanged. A natural geometric refinement is to ask whether one can define a dual representative $(\beta^\perp, g^\perp)$ that simultaneously moves in both coordinates, with connecting segment orthogonal to the crossover curve.

Crossover geometry in (β, g) . For fixed θ , define

$$y(\beta, g, \theta) = \log \chi(\beta, g, \theta), \quad \mathcal{C}_\theta := \{(\beta, g) : y = 0\}. \quad (39)$$

In one-channel scaling,

$$y(\beta, g, \theta) = 2 \log g + \log \chi_0(\beta, \theta), \quad (40)$$

so

$$\nabla y = \left(\partial_\beta \log \chi_0, \frac{2}{g} \right). \quad (41)$$

Hence the Euclidean normal direction to $\mathcal{C}_\theta$ is generated by ∇y , and the tangent slope is

$$g'_\star(\beta, \theta) = -\frac{g_\star(\beta, \theta)}{2} \partial_\beta \log \chi_0(\beta, \theta). \quad (42)$$

Definition 6 (Normal-ray dual representative). Assume $y \in C^2$ and $\nabla y \neq 0$ on $\mathcal{C}_\theta$. Let $\mathcal{U}_\theta$ be a tubular neighborhood where each $p \in \mathcal{U}_\theta$ has unique closest point $\pi(p) \in \mathcal{C}_\theta$, and write

$$p = \pi(p) + \lambda \hat{\nu}(\pi(p)), \quad (43)$$

with $\hat{\nu}$ the unit normal field. Define $p^\perp = \mathcal{D}_\perp(p)$ as the point on the same normal ray

$$p^\perp = \pi(p) + \lambda^\perp \hat{\nu}(\pi(p)), \quad (44)$$

such that

$$y(p^\perp) = -y(p). \quad (45)$$

Theorem 2 (Local existence and involution of normal duality). Under Definition 6, there exists $\delta > 0$ such that for $|\lambda| < \delta$:

1. the scalar $\lambda^\perp$ exists and is unique;
2. $\mathcal{D}_\perp$ is an involution, $\mathcal{D}_\perp^2 = \text{id}$;
3. the segment joining p and $p^\perp$ is orthogonal to $\mathcal{C}_\theta$ at $\pi(p)$.

Proof. Fix $c \in \mathcal{C}_\theta$ and define $\Phi_c(\lambda) := y(c + \lambda \hat{\nu}(c))$. Since $y(c) = 0$ and $\Phi'_c(0) = \nabla y(c) \cdot \hat{\nu}(c) = \|\nabla y(c)\| > 0$, Φ_c is strictly monotone on a small interval by continuity. Therefore, for any λ in that interval, the equation $\Phi_c(\lambda^\perp) = -\Phi_c(\lambda)$ has unique solution $\lambda^\perp$. This defines $\mathcal{D}_\perp$ locally. Applying the same construction twice gives $\Phi_c(\lambda^{\perp\perp}) = -\Phi_c(\lambda^\perp) = \Phi_c(\lambda)$, and injectivity of Φ_c yields $\lambda^{\perp\perp} = \lambda$, hence involution. Orthogonality is immediate because both points lie on one normal ray at $c = \pi(p)$. $\square$

Corollary 3 (Simultaneous (β, g) dual representative). The map $\mathcal{D}_\perp$ defines a local simultaneous dual

$$(\beta, g) \mapsto (\beta^\perp, g^\perp), \quad (46)$$

with

$$y(\beta^\perp, g^\perp, \theta) = -y(\beta, g, \theta). \quad (47)$$

Generically both coordinates change. In the special case where $\partial_\beta \log \chi_0 \approx 0$ at the crossover point, the normal is nearly vertical and the map reduces to the fixed- β duality to leading order.

Interpretation. This provides a geometrization of the strong-weak exchange: instead of viewing duality only as an algebraic inversion in g at fixed β , one may view it as local reflection across the crossover manifold along its normal bundle. The orientation-gauge Z_2 then acts as an independent discrete fiber operation on top of this base-manifold reflection.

SPIN-BOSON FIRST EXAMPLE

Role in paper. This section is an instantiation of the general framework, not the conceptual centre of the manuscript. The general theory is already complete before this example. The spin-boson model is used because it is the canonical finite-dimensional benchmark where mean-force weak/ultrastrong structure is explicit [10, 15].

Specialisation from the general framework

Model and closed parameters.

$$H_Q = \frac{\omega_q}{2} \sigma_z, \quad (48)$$

$$H_X = \sum_k \omega_k b_k^\dagger b_k, \quad (49)$$

$$f(\theta) = \cos \theta \sigma_z - \sin \theta \sigma_x, \quad (50)$$

$$B = \sum_k c_k (b_k + b_k^\dagger), \quad (51)$$

$$H_{\text{int}} = g f(\theta) \otimes B, \quad (52)$$

$$H_{\text{tot}} = H_Q + H_X + H_{\text{int}}. \quad (53)$$

Using Definition 1, the reduced equilibrium operator is partitioned as

$$\frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} = e^{-\beta H_Q/2} e^S e^{-\beta H_Q/2}, \quad (54)$$

where $S \equiv S_\beta$ is the qubit influence generator.

For a Gaussian bath, closure compresses bath dependence into $\Sigma_z(\beta, g)$ and $\Sigma_\perp(\beta, g)$ [9, 15, 16]. With $c = \cos \theta$, $s = \sin \theta$, define

$$\chi := s \sqrt{c^2 \Sigma_\perp^2 + s^2 \Sigma_z^2}, \quad (55)$$

$$\gamma(\chi) := \frac{\tanh \chi}{\chi}, \quad (56)$$

$$\Theta := \text{arctanh}(\gamma(\chi) s^2 \Sigma_z) - \frac{\beta \omega_q}{2}, \quad (57)$$

$$m_z := \tanh \Theta, \quad r := |\tanh \chi|, \quad (58)$$

$$\tan \eta := -\cot \theta \frac{\Sigma_\perp}{\Sigma_z}. \quad (59)$$

Hence radial data $(r, \mathcal{P})$ are controlled by χ , while directional data are controlled by (η, Θ) .

Single-channel scaling. With $K(u) = g^2 K_0(u)$, one has

$$\Sigma_z(\beta, g) = g^2 \Sigma_z^{(0)}(\beta), \quad \Sigma_\perp(\beta, g) = g^2 \Sigma_\perp^{(0)}(\beta), \quad (60)$$

so

$$\chi(\beta, g, \theta) = g^2 \chi_0(\beta, \theta), \quad (61)$$

$$\chi_0(\beta, \theta) := s \sqrt{c^2 (\Sigma_\perp^{(0)})^2 + s^2 (\Sigma_z^{(0)})^2}, \quad (62)$$

and

$$g_\star(\beta, \theta) = \chi_0(\beta, \theta)^{-1/2}, \quad g^\vee = \frac{g_\star^2}{g}. \quad (63)$$

Easy $g \rightarrow \infty$ route via the influence operator

Why this limit is easy. Use the symmetrized influence generator

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_\perp (\sigma_x \cos \phi_f + \sigma_y \sin \phi_f). \quad (64)$$

Let $M := S - \Delta_0 \mathbb{I}$, $M^2 = \chi^2 \mathbb{I}$. Then exactly

$$e^S = e^{\Delta_0} \cosh \chi (\mathbb{I} + \gamma M), \quad \gamma = \frac{\tanh \chi}{\chi}, \quad (65)$$

and

$$\rho_\Delta = \frac{e^S}{\text{Tr} e^S}. \quad (66)$$

No additional resummation is required: one evaluates closed functions of χ , Δ_z , $\Delta_\perp$, then takes limits.

Proposition 4 (Easy-limit statement). *At fixed (β, θ) , the $g \rightarrow \infty$ asymptotic in the spin-boson closure is directly controlled by the influence-state representation $\rho_\Delta = e^S / \text{Tr} e^S$. In particular, the radial part saturates as $r \rightarrow 1$ through $\tanh \chi$, while orientation is obtained from finite ratios encoded in $(\Delta_z, \Delta_\perp)$ through γM .*

Proof. Under single-channel scaling, $\chi = g^2 \chi_0 \rightarrow \infty$ for fixed (β, θ) with $\chi_0 > 0$. Hence $\tanh \chi \rightarrow 1$, so the radial part is explicit. Since e^S is already in closed form, the orientation is read directly from the normalized traceless part γM , whose components are finite functions of the closed-response ratios. $\square$

Mapping to the hard low-temperature sector via dual representative

Control-regime transfer. For fixed $g > 0$, low-temperature analysis is generally hard because $\chi_0(\beta, \theta)$ and response kernels become strongly structured as $\beta \rightarrow \infty$. If $\chi_0(\beta, \theta) \rightarrow \infty$, then

$$g^\vee(\beta, \theta; g) = \frac{1}{\chi_0(\beta, \theta) g} \rightarrow 0, \quad (67)$$

$$\chi(\beta, g^\vee, \theta) = \chi(\beta, g, \theta)^{-1} \rightarrow 0. \quad (68)$$

Thus the same hard strong- χ regime at (β, g) is evaluated via weak- χ expansion at $(\beta, g^\vee)$.

Corollary 4 (Hard-limit transfer). *Assume $\chi_0(\beta, \theta) \rightarrow \infty$ as $\beta \rightarrow \infty$ and Q is admissible. Then the low- T , strong- χ asymptotic of $Q(\beta, g, \theta)$ is obtained from the weak- χ asymptotic of $Q(\beta, g^\vee, \theta)$, with $g^\vee = g_\star^2/g$.*

Proof. Immediate from Theorem 1 and Proposition 2. $\square$

Canonical-channel threshold, angle dependence, and phase-diagram scope

Canonical channel ($\theta = \pi/2$): closure-exact threshold. Specialize first to $\theta = \pi/2$ ($c = 0$, $s = 1$), where the transverse channel vanishes. Then

$$\chi(\beta, g) = g^2 \Sigma_z^{(0)}(\beta; s), \quad \Theta(\beta, g) = \chi(\beta, g) - \frac{\beta \omega_q}{2}, \quad (69)$$

so

$$m_z(\beta, g; s) = \tanh \left[g^2 \Sigma_z^{(0)}(\beta; s) - \frac{\beta \omega_q}{2} \right]. \quad (70)$$

With

$$\lambda_s := \lim_{\beta \rightarrow \infty} \frac{\Sigma_z^{(0)}(\beta; s)}{\beta}, \quad (71)$$

the $T = 0$ sign-inversion threshold is

$$g_c^2(s) = \frac{\omega_q}{2\lambda_s}. \quad (72)$$

For

$$J_s(\omega) = 2g^2 \omega_c^{1-s} \omega^s e^{-\omega/\omega_c}, \quad s > 0, \quad (73)$$

this gives

$$\lambda_s = 2\omega_c \Gamma(s), \quad g_c^2(s) = \frac{\omega_q}{4\omega_c \Gamma(s)}. \quad (74)$$

This result is exact within the canonical-channel closure.

Angle-dependent threshold condition from the same closure. For generic θ , define

$$D_0(\beta, \theta) := s^2 \Sigma_z^{(0)}(\beta, \theta), \quad X_0(\beta, \theta) := \chi_0(\beta, \theta), \quad (75)$$

so $D = g^2 D_0$, $\chi = g^2 X_0$, and

$$\gamma s^2 \Sigma_z = \frac{\tanh(g^2 X_0)}{g^2 X_0} g^2 D_0 = Q_\theta(\beta) \tanh(g^2 X_0), \quad (76)$$

with

$$Q_\theta(\beta) := \frac{D_0(\beta, \theta)}{X_0(\beta, \theta)} \in (0, 1]. \quad (77)$$

The finite- β sign-change condition $\Theta = 0$ is therefore

$$Q_\theta(\beta) \tanh(g_c^2 X_0) = \tanh\left(\frac{\beta \omega_q}{2}\right). \quad (78)$$

Hence

$$g_c^2(\beta, \theta) = \frac{1}{X_0} \operatorname{arctanh} \left[\frac{\tanh(\beta \omega_q/2)}{Q_\theta(\beta)} \right], \quad (79)$$

only when $\tanh(\beta \omega_q/2) < Q_\theta(\beta)$; otherwise no finite g_c exists.

In the $T \rightarrow 0$ limit, $\tanh(\beta \omega_q/2) \rightarrow 1$, so a finite threshold requires $Q_\theta \rightarrow 1$, equivalently vanishing transverse contribution ($c \Sigma_\perp^{(0)} \rightarrow 0$). Thus, in this closure, the finite- g zero-temperature threshold is confined to the canonical axis ($\theta = \pi/2$, or other points where the transverse sector is strictly absent).

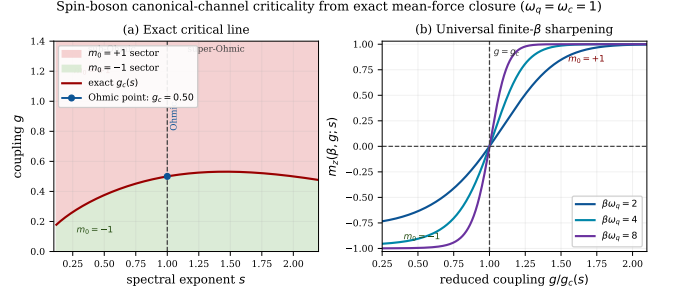


FIG. 2. Canonical-channel ($\theta = \pi/2$) closure threshold by spectral exponent. (a) $g_c(s) = \sqrt{\omega_q/(4\omega_c \Gamma(s))}$ from Eq. (74). (b) Finite- β sharpening in reduced coupling $x = g/g_c(s)$: $m_z = \tanh[(\beta \omega_q/2)(x^2 - 1)]$.

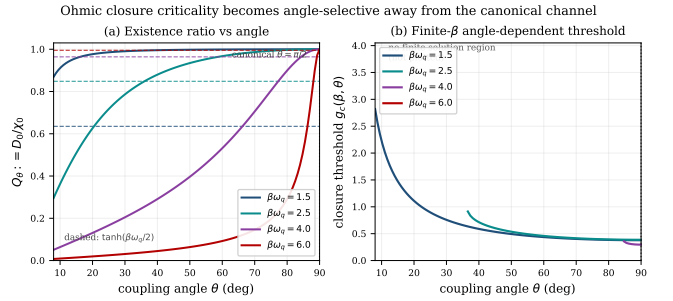


FIG. 3. Ohmic ($s = 1$) angle dependence of the closure threshold. (a) Existence ratio $Q_\theta(\beta)$ versus θ , compared with $\tanh(\beta \omega_q/2)$ (dashed). (b) Finite- β threshold $g_c(\beta, \theta)$ from Eq. (79); regions with no finite solution are omitted. As β increases, the finite-threshold corridor contracts toward $\theta = \pi/2$.

Implementation note. Figures 2 and 3 are generated by `paper_dual_hmf/validation/make_spin_boson_phase_diagram_figure.py` and `paper_dual_hmf/validation/make_spin_boson_angle_phase_figure.py`.

Interpretation and literature scope. The g_c formulas above are closure-level thresholds of the present mean-force construction, with Eq. (74) living on the canonical $\theta = \pi/2$ slice. They should not be read as a full-universality phase diagram of the spin-boson model for arbitrary θ : the established unbiased model phenomenology includes Ohmic KT criticality and no localization transition in the super-Ohmic case [10, 11]. For sub-Ohmic dissipation, NRG/QMC and analytic studies report a genuine transition with s -dependent critical behavior [12, 13, 17–19]. The angle-resolved closure map here is therefore a representative-control statement, to be benchmarked against those nonperturbative universality analyses rather than replacing them.

Dual-map analytic entanglement entropy

Closed entropy formula from the Bloch radius. For any qubit state with Bloch radius $r \in [0, 1]$, the eigenvalues are $\lambda_{\pm} = (1 \pm r)/2$, so the reduced-state von Neumann entropy is

$$S_Q = -\text{Tr}(\rho_Q \log \rho_Q) = -\sum_{\pm} \lambda_{\pm} \log \lambda_{\pm}. \quad (80)$$

Using the closure relation $r = |\tanh \chi|$, one obtains the exact analytic form

$$S_Q(\beta, g, \theta) = h_2\left(\frac{1 + |\tanh \chi(\beta, g, \theta)|}{2}\right), \quad (81)$$

where $h_2(x) := -x \log x - (1 - x) \log(1 - x)$ and $\chi(\beta, g, \theta) = g^2 \chi_0(\beta, \theta)$. In the zero-temperature limit, when the joint state approaches a pure ground state, this equals the system-bath entanglement entropy.

Proposition 5 (Dual representative entropy map). *Let $g^{\vee} = g_*^2/g$, so $\chi^{\vee} = \chi^{-1}$. Then*

$$S_Q(\beta, g^{\vee}, \theta) = h_2\left(\frac{1 + |\tanh(\chi^{-1})|}{2}\right). \quad (82)$$

Hence the strong-branch entropy at (β, g, θ) is an exact analytic function of the dual weak coordinate $\chi^{\vee}$:

$$S_Q(\beta, g, \theta) = h_2\left(\frac{1 + |\tanh((\chi^{\vee})^{-1})|}{2}\right). \quad (83)$$

Proof. Substitute $r = |\tanh \chi|$ into the qubit eigenvalue formula and use $\chi^{\vee} = \chi(\beta, g^{\vee}, \theta) = \chi(\beta, g, \theta)^{-1}$ from Theorem 1. $\square$

Asymptotic transfer for entropy. From the closed form,

$$S_Q(\chi) = \log 2 - \frac{\chi^2}{2} + O(\chi^4), \quad \chi \rightarrow 0, \quad (84)$$

and

$$S_Q(\chi) = (1 + 2\chi)e^{-2\chi} + O(e^{-4\chi}), \quad \chi \rightarrow \infty. \quad (85)$$

With $\chi^{\vee} = \chi^{-1}$, the strong-branch form is equivalently

$$S_Q = (1 + 2/\chi^{\vee})e^{-2/\chi^{\vee}} + O(e^{-4/\chi^{\vee}}), \quad \chi^{\vee} \rightarrow 0. \quad (86)$$

Support from existing analytical and numerical studies. The entropy trends encoded above are consistent with prior spin-boson analyses. NRG work found that for the unbiased Ohmic case the entanglement entropy grows with dissipation and, with bias, develops a maximum before localization [20]. Analytical scaling and field-theory treatments identified universal entanglement structure

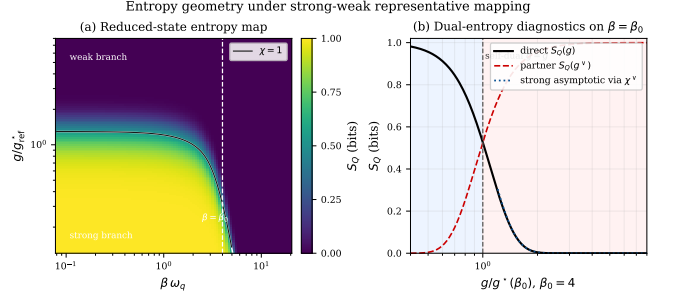


FIG. 4. Entropy diagnostics from the dual map. (a) Reduced-state entropy $S_Q/\log 2$ across (β, g) , with the crossover contour $\chi = 1$ and the $\beta = \beta_0$ slice used in panel (b). In the $\beta \rightarrow \infty$ limit (pure total ground state), this becomes the system-bath entanglement entropy. (b) Coupling sweep at fixed β_0 : direct entropy $S_Q(g)$, entropy of the dual representative $S_Q(g^{\vee})$ (showing representative non-equality), and strong-branch analytic estimate written in the dual weak coordinate $\chi^{\vee} = \chi^{-1}$; the vertical marker $g = g^*(\beta_0)$ indicates the self-dual point.

near the dissipative critical point [21, 22]. In the strong-dissipation regime, variational/multipolaron studies resolved explicit system-environment entanglement structure compatible with the entropy suppression on localized branches [23, 24]. The dual representative map presented here adds a closed analytic route that links these branch behaviors by $\chi^{\vee} = \chi^{-1}$.

Implementation note. Figure 4 is generated by `paper_dual_hmf/validation/make_spin_boson_entanglement_duality_figure.py`.

NUMERICAL VALIDATION PROTOCOL

Validation target. This section specifies a minimal validation suite aligned with the theory sections on involution, orientation gauge, and composed representative action. The goal is not broad benchmarking, but targeted falsification of the duality identities.

Single core figure specification

Implementation note. The figure is produced by `paper_dual_hmf/validation/make_core_dual_map_figure.py`, which is self-contained by design and writes outputs to `paper_dual_hmf/manuscript/figures/`.

Exact versus analytic agreement degree

Interpretation. Yes, the exact and analytic forms agree to high precision once the analysis is in the intended strong-asymptotic region. At the self-dual neighborhood ($x \approx 1$), disagreement is expected because asymptotics

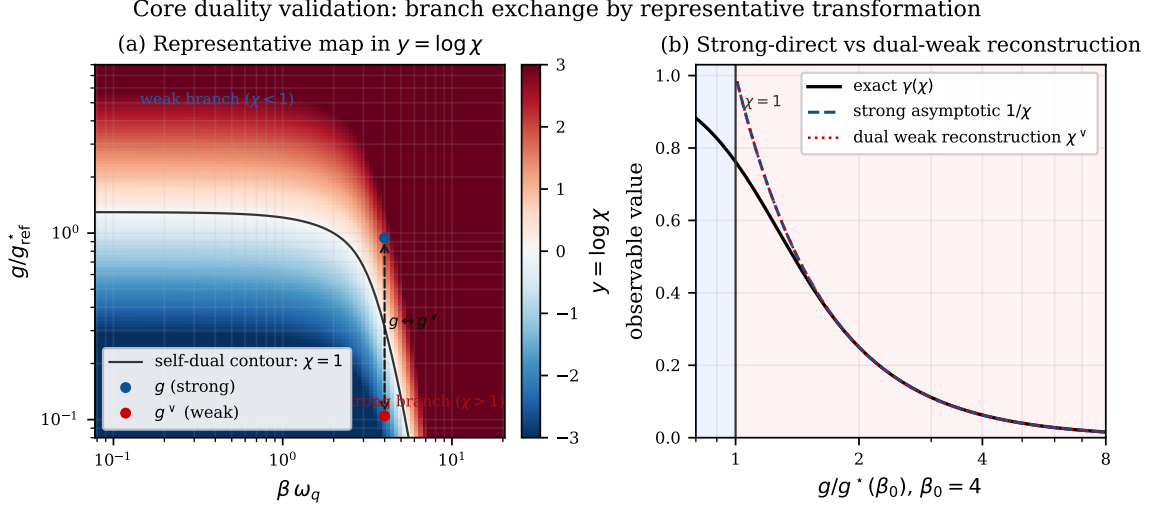


FIG. 5. Core dual-representative validation. **(a)** Diverging $y = \log \chi$ map in (β, g) , with the self-dual contour $\chi = 1$, explicit weak/strong branch labels, and a fixed- β representative pair $g \leftrightarrow g^\vee$. **(b)** Line-style decoding of asymptotic exchange at fixed β_0 : black solid = $\gamma(\chi)$ exact, blue dashed = $1/\chi$ (strong asymptotic), red dotted = $\chi^\vee = \chi(\beta_0, g^\vee, \theta)$ (dual weak variable). Convergence on the strong side ($g/g^* > 1$) visualizes the control-regime transfer claim.

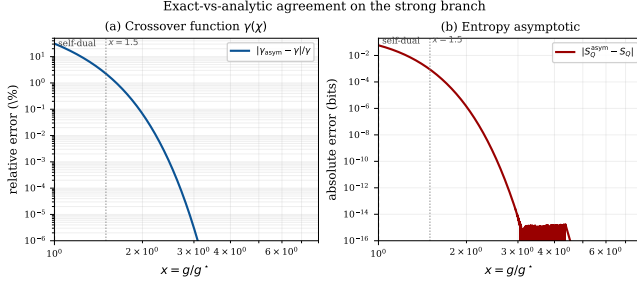


FIG. 6. Quantitative agreement diagnostics between exact closure expressions and analytic asymptotics on the strong branch $x := g/g^* \geq 1$. **(a)** Relative error of $\gamma_{\text{asym}} = 1/\chi$ against $\gamma_{\text{exact}} = \tanh \chi / \chi$ with $\chi = x^2$. **(b)** Absolute entropy error in bits for $S_Q^{\text{asym}} = ((1 + 2\chi)e^{-2\chi})/\ln 2$ against exact $S_Q = h_2((1 + |\tanh \chi|)/2)$. The dominant mismatch is localized near the crossover; errors collapse rapidly for $x \gtrsim 1.5$.

are not yet controlled. By $x \geq 1.5$, γ -errors are already at the few-percent level (max 2.24%), while entropy errors are below 10^{-3} bits. By $x \geq 2$, agreement is effectively exact for manuscript purposes: $< 10^{-3}$ relative error for γ and $O(10^{-6})$ bits for entropy.

Implementation note. Figure 6 and Table I are generated from `paper_dual_hmf/validation/make_exact_vs_analytic_comparison.py` with numeric output written to `paper_dual_hmf/validation/output/exact_vs_analytic_metrics.csv`.

TABLE I. Agreement degree in fixed strong-branch windows $x \geq x_{\min}$, with $\delta_\gamma := |\gamma_{\text{asym}} - \gamma_{\text{exact}}|/\gamma_{\text{exact}}$ and $\Delta S_Q := S_Q^{\text{asym}} - S_Q^{\text{exact}}$. Entries for δ_γ are in percent, and entries for $|\Delta S_Q|$ are in bits.

$x_{\min}$	max δ_γ	RMS δ_γ	max $ \Delta S_Q $	RMS $ \Delta S_Q $
1.0	31.30%	6.68%	5.87×10^{-2}	1.08×10^{-2}
1.2	11.87%	2.33%	1.45×10^{-2}	2.30×10^{-3}
1.5	2.24%	0.389%	8.79×10^{-4}	1.16×10^{-4}
2.0	0.0668%	0.00978%	1.37×10^{-6}	1.48×10^{-7}

Acceptance checks

1. Algebraic: $(g^\vee)^\vee = g$ and $\chi(\beta, g^\vee, \theta) = 1/\chi(\beta, g, \theta)$.
2. Orientation: $\beta'_{\text{eff}} + \beta_{\text{eff}} = 2\beta$, and radial invariants are orientation-even.
3. Asymptotic exchange: strong-branch direct evaluation agrees with dual weak-branch expansion in target limits.
4. Manuscript: compile pass with theorem/interpretation tagging complete.

Data and script reuse policy

Reuse existing crossover and involution plotting pipelines first; avoid heavy new ED campaigns in this phase.

DISCUSSION AND OUTLOOK

Sections 2–5 establish the representative-level duality structure: a strong-weak involution on the crossover coordinate and an orientation-gauge affine reflection for effective-temperature labels. Section 6 demonstrates these mechanisms in the spin-boson qubit without elevating that model to a claim of universality.

The transport mechanism requires two ingredients: (i) an explicit positive crossover coordinate with distinct weak/strong asymptotics, and (ii) a closed representative map that acts algebraically on the control labels. When ever both are present, asymptotic work can be moved between branches without changing the analytic functional family.

The proofs here are representative-level and asymptotic. They do not assert equality of physical states under parameter inversion. They also do not provide a full classification of all models admitting such discrete actions; the spin-boson section is an existence example with closed formulas.

Interpretation of the spin-boson example should remain anchored to established nonperturbative results: Ohmic KT behavior and super-Ohmic delocalization are standard benchmarks, while sub-Ohmic criticality is supported by NRG/QMC and complementary analytic approaches [10–13, 17–19]. In that context, the present contribution is a representative duality geometry for mean-force calculations, not a replacement for full universality classification.

Natural next testbeds are multi-channel models, where multiple χ_a coordinates may generate a $(Z_2)^k$ reflection structure, and finite- N systems with non-trivial eigen-label symmetries. The minimal numerical protocol in Section 7 is designed to scale to those settings with only modest changes.

Proof Skeletons and Derivation Bookkeeping

Proof sketch: strong-weak involution theorem

With $\chi = g^2 \chi_0$, $\chi_0 > 0$, define $g^\vee = 1/(\chi_0 g)$. Then

$$\chi^\vee = \chi_0 (g^\vee)^2 = \frac{1}{\chi_0 g^2} = \chi^{-1}. \quad (87)$$

Applying the map twice gives

$$(g^\vee)^\vee = \frac{1}{\chi_0 g^\vee} = g. \quad (88)$$

Hence $y = \log \chi$ reflects as $y^\vee = -y$.

Proof sketch: affine reflection lemma

Let $\beta_{\text{eff}} = \beta - \delta\beta$, with

$$\delta\beta = \frac{2}{\omega_q} \operatorname{arctanh}(u). \quad (89)$$

Under orientation gauge, $u \mapsto -u$, so

$$\delta\beta' = -\delta\beta. \quad (90)$$

Therefore

$$\beta'_{\text{eff}} = \beta - \delta\beta' = \beta + \delta\beta = 2\beta - \beta_{\text{eff}}. \quad (91)$$

Proof sketch: even/odd observable split

The gauge map acts as $(\chi, u) \mapsto (\chi, -u)$. For $O_{\text{rad}} = F(\chi)$, $O'_{\text{rad}} = F(\chi) = O_{\text{rad}}$. For odd $O_{\text{odd}} = H(\chi, u)$ with $H(\chi, -u) = -H(\chi, u)$, one has $O'_{\text{odd}} = -O_{\text{odd}}$.

Equation bookkeeping index

Maintain one-to-one mapping with
paper_dual_hmf/notes/equation_registry.md.

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